Blessed G/Michael Catholic Secondary School

1st Semester Biology Test two for Grade 11

Dec, 2018 E.C Time Allowed -40min'

I. Write true if the statement is correct or write false if it is not correct

True 1. The active site of an enzyme fits complementarily to the substrate during action.

True 2. Our body uses L-amino acids and L- Hexose sugars to build polymers.

True 3. Sensitization is a process of amplifying a response.

True 4. Reproductive behaviour is a comprehensive behaviour which includes also Courtship behaviour.

False 5. Interspecific competition always includes competition for resource and mate.

II. Match the types of Enzymes in column B With the type of enzyme specificity in column A

A

C 1. Bond specificity

B 2. Group specificity

D 3. Substrate specificity

E 4. Cofactor specificity

A 5. Optical specificity

B

A. Gluco kinase -Optical

B. Pepsin Group

C. Lipase Bond

D. Lactase substrate

E. NAD dehydrogenase co-factor

III. Choose the best answer from the given alternatives and write in Capital letters only.

D 1. Which of these statements about enzymes is true?

A. They are proteins with unique 3D structure. C. Their main action is to reduce activation energy

B. They do not alter the overall free energy change. D. All of these

A 2. Which of the following property of enzymes prevent them from passing through semipermeable membranes?

A. Colloidal B. Solubility C. Precipitation D. Catalytic

D 3. Which of the following is false about primary structure of proteins? It is :-

A. Determined by our genes C. Determines the nature of secondary structure

B. The sequence of amino acids D. Formed by ionic bonds among R-groups.

B 4. Which of the following is false about secondary structure of proteins?

A. It exists in two variants. C. It is determines the shape of the tertiary structure.

B. It is determined by our genes D. Formed by H- bonds between C=O and H-N- groups.

A 5. Which of the following is false about Quaternary structure of proteins?

A. It occurs in many enzymes. C. Haemoglobin is an example of it.

B. Formed by H- bonds between sub units. D. It is the most complex protein structure.

6. Eusocial behaviour is different from other social behaviours due to the presence of all except:-

A. Social caste C. Rearing infants together.

B. Division of lever D. Living to gather

B 7. Negative feedback in homeostasis refers to a :

A. response that amplifies the initial stimulus C. system regulated only by the central nervous system.

B. response that reduces deviation from a set point. D. one-time corrective action that does not repeat.

B 8. Antidiuretic hormone (ADH) acts mainly on:

A. Proximal tubule to increase sodium reabsorption C. Loop of Henle to increase urea secretion

B. Collecting ducts to increase water permeability D. Distal tubule to inhibit potassium secretion

B 9. If you play in the snow, your body maintains a constant core temperature by …

A. Vasodilation and sweating. C. Sweating and shivering

B. Vasoconstriction and shivering. D. Vasoconstriction and vasodilation

A 10. Which one is not the type of adaptation of behaviour needed in hot desert places by mammals?

A. Decreasing S.A/V C. Water bathing

B. Aestivation D. Decreasing hairiness.

11. Which of the following best explains the role of the hypothalamus in homeostasis?

B A. It only controls circadian rhythms and not metabolic processes
B. It integrates autonomic, endocrine, and behavioural responses to maintain internal condition.
C. It is only responsible for long-term hormonal adjustments via the thyroid
D. It functions exclusively in voluntary motor control.

C 12. Negative feedback is expressed all except it is:-
A. the way the body maintains homeostasis C. a response that enhances a stimulus.
B. a response that opposes a stimulus. D. a response that maintains a dynamic state of equilibrium

A 13. Which one is not the signal or real displays of the corresponding the following territorial animal?
A. Young Female Wolfs- Urine demarking C. Lions –Roaring
B. Robin- Showing red breast D. Human- Making Metal fence

C 14. Which of the following is a correct definition of "positive feedback"?
A. The process by which the body maintains homeostasis
B. A mechanism in which the body's response to a stimulus opposes the stimulus
C. A mechanism whereby the body responds to a stimulus by acting to enhance the stimulus
D. The dynamic equilibrium maintained by an integrating centre which causes an effector to respond to the stimulus received by the receptor.

B 15. How is the distance to a food source communicated by a dancing honeybee?
A. by the direction it waggles its abdomen
B. by how far it moves during the straight run portion of the dance
C. by which direction it turns after making the straight run
D. by the tempo or degree of vigor of the dance

D 16. Courtship behaviors in animals involves:-
A. Visual cues B. Auditory cues C. Chemical signals D. All of the above are involved.

B 17. Color or shape in an animal's covering that helps it blend into its environment is called
A. Mimicry B. Camouflage C. Behavior D. Instinct

D 18. Which of the following protein structures is destroyed by denaturation?
A. Primary B. Secondary C. Tertiary D. Both B and C

B 19. Which one of the following Caste of honey bee has haploid chromosomes?
A. Queen B. Drone C. Workers D. All of these

D 20. True social behavior is present in:-
A. Termites B. Ants and beetles C. Humans and Gorilla D. Termites, ant and Bees

B 21. Soldier caste is present in [illegible]:-
A. House flies B. Termites C. Lion D. Humans

A 22. The group of animals which are led by queens only include all except:-
A. Humans B. Honey bees C. Hyena D. Termites

Name Nahomin Menale Section 11 C No 33 29

Part I
1. True
2. True
3. True
4. True
5. False

Part II
1. C
2. B
3. D
4. E
5. A

Part III
1. D
2. A
3. D
4. B
5. A
6. [illegible]
7. B
8. B
9. B
10. A
11. B
12. C
13. A
14. C
15. B
16. D
17. B
18. D
19. B
20. D
21. B
22. A